

DBMS - SQL Queries & DML

1. Which SQL command is used to retrieve data from a database?
 - A) GET
 - B) SELECT
 - C) FETCH
 - D) RETRIEVE

2. Which DML command is used to remove all records from a table while keeping the table structure?
 - A) DELETE
 - B) DROP
 - C) TRUNCATE
 - D) REMOVE

3. What is the correct SQL syntax to insert a new row into a table named 'Students'?
 - A) INSERT INTO Students VALUES (1, 'John');
 - B) ADD ROW TO Students (1, 'John');
 - C) INSERT Students (1, 'John');
 - D) INPUT INTO Students (1, 'John');

4. Which clause is used to filter the results of a SELECT query based on a condition?
 - A) ORDER BY
 - B) GROUP BY
 - C) HAVING
 - D) WHERE

5. Which keyword is used to eliminate duplicate rows in a SELECT query result?
 - A) UNIQUE
 - B) DISTINCT
 - C) DIFFERENT
 - D) SINGLE

6. Which operator is used to search for a pattern in a column?
 - A) LIKE
 - B) MATCH
 - C) SEARCH
 - D) AS

7. Which SQL command is used to modify existing records in a database table?
 - A) MODIFY
 - B) CHANGE
 - C) UPDATE
 - D) ALTER

8. What is the output of the query: SELECT COUNT(*) FROM Employees;?
 - A) The sum of all numeric columns
 - B) The number of rows in the table
 - C) The number of columns in the table
 - D) The first record of the table

9. Which clause is used to sort the result set in ascending or descending order?

- A) SORT BY
- B) ALIGN BY
- C) ORDER BY
- D) SEQUENCE BY

10. Which aggregate function is used to calculate the average value of a numeric column?

- A) AVG()
- B) MEAN()
- C) SUM()
- D) TOTAL()

11. Which SQL command is used to permanently save changes made during the current transaction?

- A) SAVE
- B) COMMIT
- C) ROLLBACK
- D) GRANT

12. What will the query "SELECT * FROM Students WHERE Name LIKE 'A%'" return?

- A) Names ending with A
- B) Names containing A
- C) Names starting with A
- D) Names that are exactly A

13. Which command is used to undo changes in a transaction before they are committed?

- A) UNDO
- B) REVERT
- C) ROLLBACK
- D) CANCEL

14. Which clause is used to group rows that have the same values into summary rows?

- A) GROUP BY
- B) PARTITION BY
- C) CLUSTER BY
- D) SORT BY

15. What is the wildcard character in SQL used to represent zero or more characters?

- A) ?
- B) #
- C) *
- D) %

16. Which of the following is a Data Manipulation Language (DML) command?

- A) CREATE
- B) SELECT
- C) ALTER
- D) DROP

17. The HAVING clause is typically used with which of the following?

- A) SELECT
- B) WHERE
- C) GROUP BY
- D) ORDER BY

18. Which operator is used to specify a range of values in a WHERE clause?

- A) IN
 - B) BETWEEN
 - C) RANGE
 - D) LIMIT
-

19. Which function is used to find the highest value in a specific column?

- A) HIGH()
 - B) TOP()
 - C) MAX()
 - D) LARGEST()
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20. What happens when you use the DELETE command without a WHERE clause?

- A) Only the last row is deleted
 - B) All rows in the table are deleted
 - C) An error occurs
 - D) The table structure is deleted
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Answer Key — DBMS - SQL Queries & DML

1. B	2. C	3. A	4. D	5. B
6. A	7. C	8. B	9. C	10. A
11. B	12. C	13. C	14. A	15. D
16. B	17. C	18. B	19. C	20. B

Target1st

DBMS - Normalization & Keys

1. Which key is a minimal set of attributes that can uniquely identify a record in a table?
 - A) Primary Key
 - B) Super Key
 - C) Candidate Key
 - D) Foreign Key

2. What is the primary purpose of normalization in a database?
 - A) To increase storage space
 - B) To reduce data redundancy
 - C) To improve query speed
 - D) To encrypt data

3. Which normal form deals with the removal of partial functional dependencies?
 - A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF

4. What is a Foreign Key used for?
 - A) Uniquely identifying a row
 - B) Linking two tables together
 - C) Sorting the data
 - D) Creating a backup

5. A table is in 3NF if it is in 2NF and has no:
 - A) Partial dependencies
 - B) Transitive dependencies
 - C) Multi-valued dependencies
 - D) Join dependencies

6. Which key consists of more than one attribute to uniquely identify a record?
 - A) Simple key
 - B) Composite key
 - C) Atomic key
 - D) Single key

7. What is the highest normal form required for most business applications to avoid anomalies?
 - A) 1NF
 - B) 2NF
 - C) 3NF
 - D) 4NF

8. Which database anomaly occurs when we cannot insert data because some other data is missing?
 - A) Update anomaly
 - B) Deletion anomaly
 - C) Insertion anomaly
 - D) Retrieval anomaly

9. A Candidate Key that is selected to uniquely identify rows in a table is called:
 - A) Primary Key
 - B) Alternate Key
 - C) Foreign Key
 - D) Super Key

10. If an attribute in a table depends on a non-key attribute, it is called a:

- A) Partial dependency
- B) Transitive dependency
- C) Full functional dependency
- D) Multi-valued dependency

11. Which normal form requires that a table must have no multi-valued dependencies?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 4NF

12. A table is in 1NF if all its attributes are:

- A) Atomic
- B) Calculated
- C) Redundant
- D) Null

13. Which of the following is a type of key that can contain NULL values?

- A) Primary Key
- B) Unique Key
- C) Foreign Key
- D) Candidate Key

14. An "Alternate Key" is:

- A) A key used for backups
- B) Any candidate key not chosen as primary key
- C) A foreign key
- D) A primary key of another table

15. What is the result of denormalization?

- A) Improved data integrity
- B) Reduced redundancy
- C) Faster read performance
- D) No duplicate data

16. The condition where a Primary Key of one table is used in another table is known as:

- A) Super Key
- B) Foreign Key
- C) Candidate Key
- D) Composite Key

17. Which dependency is removed when moving from 1NF to 2NF?

- A) Transitive dependency
- B) Partial dependency
- C) Functional dependency
- D) Total dependency

18. A super key that has no proper subset that is also a super key is called:

- A) Candidate Key
- B) Primary Key
- C) Foreign Key
- D) Secondary Key

19. Which normal form is considered a stricter version of 3NF?

- A) 1NF
 - B) 2NF
 - C) BCNF
 - D) 5NF
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20. In a database, a 'Composite Key' is:

- A) Composed of two or more attributes
 - B) A key that does not allow duplicates
 - C) A key that is not indexed
 - D) A key that references another table
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Answer Key — DBMS - Normalization & Keys

1. C	2. B	3. B	4. B	5. B
6. B	7. C	8. C	9. A	10. B
11. D	12. A	13. C	14. B	15. C
16. B	17. B	18. A	19. C	20. A

Target1st

DBMS - Transactions & Concurrency Control

1. Which ACID property ensures that a transaction is either completed in full or not at all?
 - A) Atomicity
 - B) Consistency
 - C) Isolation
 - D) Durability

2. Which term describes the phenomenon where a transaction reads data that has been modified by another uncommitted transaction?
 - A) Lost Update
 - B) Unrepeatable Read
 - C) Dirty Read
 - D) Phantom Read

3. What is the main objective of Concurrency Control in DBMS?
 - A) To ensure data redundancy
 - B) To allow multiple users to access the database simultaneously without conflict
 - C) To reduce the size of the database
 - D) To improve database security

4. In the Two-Phase Locking (2PL) protocol, what happens during the shrinking phase?
 - A) Locks are acquired
 - B) Locks are released
 - C) No locks can be acquired or released
 - D) Only data reading is allowed

5. Which state does a transaction enter immediately after it has finished executing its final statement?
 - A) Active
 - B) Partially Committed
 - C) Committed
 - D) Failed

6. Which protocol is commonly used to prevent deadlocks in transactions by assigning timestamps?
 - A) Wait-Die
 - B) Two-Phase Commit
 - C) Strict 2PL
 - D) Log-based recovery

7. What is the primary purpose of the 'Durability' property in ACID?
 - A) To ensure data integrity after a system crash
 - B) To keep data hidden from other users
 - C) To maintain data consistency
 - D) To allow rollback of transactions

8. Which database component is responsible for recording all changes to facilitate recovery?
 - A) Data Dictionary
 - B) Transaction Log
 - C) Buffer Manager
 - D) Query Optimizer

9. What happens in a 'Cascading Rollback'?

- A) A single transaction failure leads to the rollback of many other dependent transactions
- B) A transaction is committed twice
- C) A deadlock is automatically resolved
- D) Data is permanently saved to disk

10. Which concurrency control technique uses a system-wide clock or counter to order transactions?

- A) Locking-based
- B) Timestamp-based
- C) Optimistic
- D) Multiversion

11. A transaction that is waiting for a resource held by another transaction, which is in turn waiting for the first, is in a state of:

- A) Starvation
- B) Deadlock
- C) Livelock
- D) Serialization

12. Which isolation level provides the highest degree of consistency by preventing all concurrency phenomena?

- A) Read Uncommitted
- B) Read Committed
- C) Repeatable Read
- D) Serializable

13. In 'Optimistic Concurrency Control', what is the final stage?

- A) Read Phase
- B) Validation Phase
- C) Write Phase
- D) Commit Phase

14. What does a 'Shared Lock' (S-lock) allow other transactions to do?

- A) Only read the data
- B) Write the data
- C) Both read and write
- D) Neither read nor write

15. Which ACID property ensures that the database remains in a valid state before and after the transaction?

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability

16. If a transaction fails during execution, which command is typically triggered to undo its effects?

- A) ROLLBACK
- B) COMMIT
- C) GRANT
- D) REVOKE

17. A 'Write Lock' (X-lock) on a data item prevents other transactions from:

- A) Reading only
- B) Writing only
- C) Both reading and writing
- D) Releasing locks

18. Which problem occurs when a transaction reads the same row twice and finds different data because another transaction modified it?

- A) Dirty Read
 - B) Phantom Read
 - C) Unrepeatable Read
 - D) Lost Update
-

19. Which of the following is NOT an ACID property?

- A) Atomicity
 - B) Consistency
 - C) Isolation
 - D) Authenticity
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20. In the context of Concurrency Control, what is a schedule that produces the same result as a serial execution called?

- A) Serializable
 - B) Concurrent
 - C) Atomic
 - D) Isolated
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Answer Key — DBMS - Transactions & Concurrency Control

1. A	2. C	3. B	4. B	5. B
6. A	7. A	8. B	9. A	10. B
11. B	12. D	13. B	14. A	15. B
16. A	17. C	18. C	19. D	20. A

Target1st

DBMS - Indexing & Storage

1. What is the primary purpose of creating an index on a database table?
 - A) To reduce disk space
 - B) To speed up data retrieval
 - C) To delete duplicate rows
 - D) To encrypt the data

2. Which data structure is most commonly used in database systems for indexing?
 - A) B-Tree
 - B) Stack
 - C) Queue
 - D) Linked List

3. A clustered index determines the _____ of data rows in the table.
 - A) Logical order
 - B) Physical order
 - C) Encryption level
 - D) Security access

4. How many clustered indexes can a single table have?
 - A) Multiple
 - B) Unlimited
 - C) Only one
 - D) Zero

5. What is the main disadvantage of creating too many indexes on a table?
 - A) Increased search speed
 - B) Increased security risk
 - C) Slower data modification (INSERT/UPDATE/DELETE)
 - D) Higher database normalization

6. A non-clustered index contains:
 - A) The actual data rows
 - B) Only the index key and a pointer to the data row
 - C) A copy of the entire table
 - D) Metadata about the table structure

7. In a B+ Tree, where are all the actual data records stored?
 - A) Only in the root node
 - B) In the internal nodes
 - C) In the leaf nodes
 - D) In the parent nodes

8. Which type of indexing is created automatically by the database when a Primary Key is defined?
 - A) Secondary index
 - B) Clustered index
 - C) Hash index
 - D) Bitmap index

9. What does the "fill factor" setting in an index represent?
 - A) Percentage of free space in index pages
 - B) Number of rows per block
 - C) Total size of the index
 - D) The number of columns indexed

10. A dense index is one where:

- A) Every search key value has an index entry
- B) Only some search key values have entries
- C) The index is larger than the table
- D) There is no pointer to the data

11. Which indexing technique uses a calculation to map a key directly to a specific disk location?

- A) B-Tree indexing
- B) Hash indexing
- C) Clustered indexing
- D) Range indexing

12. A "Sparse Index" is typically used when:

- A) Every record is indexed
- B) The file is sorted
- C) The file is unsorted
- D) The data is stored in memory

13. Which of the following is an example of an external storage device?

- A) RAM
- B) Cache
- C) Magnetic Disk
- D) CPU Register

14. In database storage, what is a "block" or "page"?

- A) The smallest unit of data transfer between disk and memory
- B) The entire database file
- C) A single row of a table
- D) A software query tool

15. What is the primary goal of disk scheduling algorithms in a DBMS?

- A) To increase query complexity
- B) To minimize disk seek time
- C) To store more data
- D) To remove redundant keys

16. Which of the following is considered "Volatile" storage?

- A) Hard Drive
- B) RAM
- C) Flash Drive
- D) Magnetic Tape

17. What is the benefit of a B+ Tree over a standard B-Tree for database storage?

- A) Easier to implement
- B) Faster sequential access
- C) Smaller root node
- D) Requires more memory

18. An index entry pointing to a physical address is known as:

- A) Record ID (RID)
- B) Table ID
- C) View ID
- D) Schema ID

19. Which operation is most affected by a poorly chosen index?

- A) SELECT
 - B) DROP
 - C) INSERT
 - D) TRUNCATE
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20. A covering index is an index that:

- A) Contains all columns requested by a specific query
 - B) Covers the entire database
 - C) Is used only for security
 - D) Replaces the primary key
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Answer Key — DBMS - Indexing & Storage

1. B	2. A	3. B	4. C	5. C
6. B	7. C	8. B	9. A	10. A
11. B	12. B	13. C	14. A	15. B
16. B	17. B	18. A	19. C	20. A

Target1st

DBMS - ER Modeling & Database Design

1. In an ER diagram, which shape is used to represent an Entity type?
 - A) Rectangle
 - B) Ellipse
 - C) Diamond
 - D) Circle
2. Which symbol is used to represent an attribute in an ER diagram?
 - A) Rectangle
 - B) Ellipse
 - C) Diamond
 - D) Square
3. A relationship between entities of the same type is known as?
 - A) Recursive relationship
 - B) Ternary relationship
 - C) Strong relationship
 - D) Mandatory relationship
4. The maximum number of relationship instances that an entity can participate in is called?
 - A) Cardinality
 - B) Modality
 - C) Degree
 - D) Attribute
5. What does a double rectangle represent in an ER diagram?
 - A) Weak entity set
 - B) Multi-valued attribute
 - C) Derived attribute
 - D) Total participation
6. An attribute that can be divided into smaller sub-parts is called a?
 - A) Composite attribute
 - B) Simple attribute
 - C) Key attribute
 - D) Derived attribute
7. A relationship involving three different entity types is called?
 - A) Ternary relationship
 - B) Binary relationship
 - C) Unary relationship
 - D) Quaternary relationship
8. What is the default cardinality of an entity in a 1:N relationship?
 - A) Many
 - B) One
 - C) Zero
 - D) Optional
9. Which attribute in an ER diagram is represented by a dashed ellipse?
 - A) Derived attribute
 - B) Multi-valued attribute
 - C) Key attribute
 - D) Weak entity

10. The property that specifies the minimum number of entity occurrences that can be involved in a relationship is?

- A) Participation constraint
- B) Cardinality
- C) Degree
- D) Entity integrity

11. Which of the following describes the number of entities participating in a relationship?

- A) Degree
- B) Cardinality
- C) Domain
- D) Tuple

12. A "Weak Entity" is one that?

- A) Does not have a primary key of its own
- B) Has only one attribute
- C) Cannot be related to other entities
- D) Is always optional

13. If an attribute can hold more than one value for a single entity, it is called a?

- A) Multi-valued attribute
- B) Composite attribute
- C) Derived attribute
- D) Simple attribute

14. Which symbol is used to represent a relationship between entities in an ER model?

- A) Diamond
- B) Rectangle
- C) Triangle
- D) Hexagon

15. A relationship where every instance of an entity must participate is known as?

- A) Total participation
- B) Partial participation
- C) Optional participation
- D) Mandatory attribute

16. The process of defining a set of sub-entities from a super-entity is called?

- A) Specialization
- B) Generalization
- C) Normalization
- D) Aggregation

17. What does the term "Generalization" represent in ER modeling?

- A) Bottom-up approach
- B) Top-down approach
- C) Data fragmentation
- D) Constraint removal

18. Which concept allows a relationship to be treated as an entity set for the purpose of further relations?

- A) Aggregation
- B) Decomposition
- C) Inheritance
- D) Union

19. An entity set that depends on the existence of another entity set is called?

- A) Weak entity set
 - B) Strong entity set
 - C) Dependent set
 - D) Independent set
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20. In ER modeling, the attribute that is computed from other attributes is known as?

- A) Derived attribute
 - B) Multi-valued attribute
 - C) Primary attribute
 - D) Composite attribute
-

Answer Key — DBMS - ER Modeling & Database Design

1. A	2. B	3. A	4. A	5. A
6. A	7. A	8. B	9. A	10. A
11. A	12. A	13. A	14. A	15. A
16. A	17. A	18. A	19. A	20. A

Target1st

Operating System - Process Management & Scheduling

1. What is a program in execution called?
 - A) Process
 - B) Thread
 - C) Task
 - D) Instruction

2. Which state is a process in when it is waiting for an event like I/O completion?
 - A) Running
 - B) Ready
 - C) Blocked
 - D) Terminated

3. Which data structure is used by the OS to store information about a process?
 - A) Process Control Block (PCB)
 - B) Stack
 - C) Queue
 - D) Link List

4. What is the process of switching the CPU from one process to another called?
 - A) Context switching
 - B) Scheduling
 - C) Spooling
 - D) Multithreading

5. Which scheduling algorithm selects the process that arrived first?
 - A) Shortest Job First
 - B) First-Come, First-Served
 - C) Priority Scheduling
 - D) Round Robin

6. Which algorithm is best suited for time-sharing systems?
 - A) FCFS
 - B) Round Robin
 - C) SJF
 - D) Priority

7. What is the primary purpose of a CPU scheduler?
 - A) To increase I/O speed
 - B) To allocate the CPU to processes
 - C) To manage memory
 - D) To prevent deadlocks

8. In a process lifecycle, what state follows the 'New' state?
 - A) Running
 - B) Waiting
 - C) Ready
 - D) Terminated

9. Which scheduling algorithm is non-preemptive by nature?
 - A) Round Robin
 - B) Shortest Job First (SJF)
 - C) Preemptive Priority
 - D) Multilevel Queue

10. What is the phenomenon where a process is waiting for resources held by another process?

- A) Starvation
- B) Deadlock
- C) Thrashing
- D) Context Switching

11. Which term describes the total time taken from the submission of a process to its completion?

- A) Turnaround time
- B) Waiting time
- C) Response time
- D) Throughput

12. What is the name of the process that is currently being executed by the CPU?

- A) Ready process
- B) Running process
- C) Blocked process
- D) Suspended process

13. Which of the following is a preemptive scheduling algorithm?

- A) FCFS
- B) SJF (Non-preemptive)
- C) Round Robin
- D) Priority (Non-preemptive)

14. What does the 'Ready Queue' store?

- A) Processes waiting for I/O
- B) Processes ready to be executed
- C) Processes that are finished
- D) Processes waiting for memory

15. What is the dispatcher responsible for?

- A) Creating a new process
- B) Giving control of the CPU to the process selected by the scheduler
- C) Managing disk space
- D) Compiling code

16. Which of the following is considered a "lightweight" process?

- A) Thread
- B) PCB
- C) Semaphore
- D) Monitor

17. What is the term for a situation where low-priority processes never get CPU time?

- A) Deadlock
- B) Aging
- C) Starvation
- D) Fragmentation

18. What is the technique used to prevent starvation by increasing the priority of processes over time?

- A) Aging
- B) Swapping
- C) Paging
- D) Buffering

19. Which scheduler decides which processes are brought into the main memory from the disk?

- A) Long-term scheduler
 - B) Short-term scheduler
 - C) Medium-term scheduler
 - D) Dispatcher
-

20. In Round Robin scheduling, what is the assigned time interval called?

- A) Time slot
 - B) Time quantum
 - C) Context slice
 - D) Response slice
-

Answer Key — Operating System - Process Management & Scheduling

1. A	2. C	3. A	4. A	5. B
6. B	7. B	8. C	9. B	10. B
11. A	12. B	13. C	14. B	15. B
16. A	17. C	18. A	19. A	20. B

Target1st

Operating System - Memory Management & Paging

1. Which memory management technique divides memory into fixed-size blocks called frames?
 - A) Segmentation
 - B) Paging
 - C) Swapping
 - D) Overlays

2. What is the logical address space of a process divided into?
 - A) Frames
 - B) Blocks
 - C) Pages
 - D) Sectors

3. The mapping from logical address to physical address is performed by which unit?
 - A) ALU
 - B) MMU
 - C) CU
 - D) Register

4. What is the primary purpose of using a Page Table in an operating system?
 - A) To store process IDs
 - B) To translate logical addresses to physical addresses
 - C) To allocate CPU cycles
 - D) To manage disk space

5. What is the situation called when a process requests a page not currently in main memory?
 - A) Page fault
 - B) Segmentation fault
 - C) Thrashing
 - D) Cache miss

6. Which memory management scheme allows the physical address space of a process to be non-contiguous?
 - A) Partitioning
 - B) Contiguous allocation
 - C) Paging
 - D) Stacking

7. Which register holds the base address of the page table?
 - A) Stack pointer
 - B) Program counter
 - C) Page-table base register
 - D) Index register

8. What is the term for the time spent waiting for a page to be swapped from disk to memory?
 - A) Latency
 - B) Access time
 - C) Page fault service time
 - D) Seek time

9. Which algorithm replaces the page that will not be used for the longest period of time?
 - A) FIFO
 - B) Optimal
 - C) LRU
 - D) LFU

10. What phenomenon occurs when the system spends more time paging than executing processes?

- A) Fragmentation
- B) Deadlock
- C) Thrashing
- D) Swapping

11. Fragmentation that occurs when memory is divided into fixed-size partitions is called?

- A) External fragmentation
- B) Internal fragmentation
- C) Page fragmentation
- D) Segment fragmentation

12. What does TLB stand for in the context of memory management?

- A) Translation Lookaside Buffer
- B) Transfer Level Block
- C) Table Lookup Buffer
- D) Total Logical Block

13. Which memory management technique uses variable-sized partitions based on logical units?

- A) Paging
- B) Segmentation
- C) Swapping
- D) Buffering

14. In a system with 4KB pages, what is the page offset size in bits?

- A) 10 bits
- B) 12 bits
- C) 16 bits
- D) 8 bits

15. Which of the following is an advantage of Paging?

- A) No external fragmentation
- B) No internal fragmentation
- C) Fast disk access
- D) Unlimited memory

16. If a process is 10 KB and the page size is 4 KB, how many pages are required?

- A) 2
- B) 3
- C) 4
- D) 5

17. What is the technique of moving processes from main memory to disk to free up space called?

- A) Paging
- B) Segmentation
- C) Swapping
- D) Thrashing

18. Which page replacement algorithm replaces the page that has been in memory the longest?

- A) LRU
- B) Optimal
- C) FIFO
- D) Random

19. What is the size of a page usually equal to?

- A) Size of a frame
 - B) Size of a segment
 - C) Size of the disk block
 - D) Size of the register
-

20. In segmentation, each entry in the segment table contains?

- A) Base and Limit
 - B) Page number and Offset
 - C) Frame number and bit
 - D) PID and Priority
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Answer Key — Operating System - Memory Management & Paging

1. B	2. C	3. B	4. B	5. A
6. C	7. C	8. C	9. B	10. C
11. B	12. A	13. B	14. B	15. A
16. B	17. C	18. C	19. A	20. A

Target1st

Operating System - Deadlocks & Synchronization

1. Which of the following is a necessary condition for a deadlock to occur?
 - A) Mutual Exclusion
 - B) Time sharing
 - C) Multiprocessing
 - D) Virtual memory

2. In the context of the Banker's algorithm, what does it primarily determine?
 - A) Memory allocation
 - B) CPU scheduling
 - C) Deadlock avoidance
 - D) Disk fragmentation

3. Which of the following conditions implies that a resource can only be released voluntarily by the process holding it?
 - A) Hold and Wait
 - B) No preemption
 - C) Circular wait
 - D) Mutual exclusion

4. A semaphore is a variable used for which purpose in an operating system?
 - A) Memory management
 - B) File access
 - C) Process synchronization
 - D) Interrupt handling

5. Which operation is typically performed by a process to indicate it is entering a critical section?
 - A) Wait()
 - B) Signal()
 - C) Release()
 - D) Free()

6. What is the main goal of a critical section?
 - A) Increase throughput
 - B) Prevent race conditions
 - C) Reduce context switching
 - D) Manage virtual memory

7. Which of the following is a synchronization tool that acts as a lock?
 - A) Mutex
 - B) Stack
 - C) Pointer
 - D) Interrupt

8. When multiple processes access and manipulate the same data concurrently, the outcome depends on the order of execution; this is called?
 - A) Race condition
 - B) Deadlock
 - C) Starvation
 - D) Paging

9. Which state is a process in if it is blocked waiting for a resource that will never become available?
- A) Running
 - B) Deadlocked
 - C) Ready
 - D) Terminated
-
10. Which algorithm is used to detect deadlocks in systems with multiple instances of each resource type?
- A) Banker's algorithm
 - B) Wait-for graph
 - C) Round Robin
 - D) First-come-first-served
-
11. What is the term for the requirement that only one process can use a shared resource at any given time?
- A) Mutual exclusion
 - B) Progress
 - C) Bounded waiting
 - D) Hold and wait
-
12. In a 'Wait-for' graph, what does a cycle indicate?
- A) High CPU usage
 - B) Deadlock
 - C) Memory overflow
 - D) I/O completion
-
13. Which of the following is a solution to the Dining Philosophers problem to prevent deadlock?
- A) Allow all to pick up the right fork
 - B) Allow one to pick up both forks simultaneously
 - C) Force philosophers to pick up an odd-numbered fork first
 - D) Restrict the number of philosophers allowed at the table
-
14. A monitor is a high-level synchronization construct that ensures what?
- A) Only one process is active within the monitor at a time
 - B) Multiple processes can access shared data simultaneously
 - C) Deadlock is physically impossible
 - D) Interrupts are disabled during execution
-
15. Which of the following is NOT a method for handling deadlocks?
- A) Deadlock prevention
 - B) Deadlock avoidance
 - C) Deadlock detection and recovery
 - D) Deadlock virtualization
-
16. What does the Signal() operation on a semaphore do?
- A) Increments the semaphore value
 - B) Decrements the semaphore value
 - C) Blocks the process
 - D) Terminates the process
-
17. What is the "Bounded Waiting" criteria in critical section problems?
- A) A limit on the number of processes
 - B) A limit on the number of times a process can enter the section
 - C) A limit on the number of times other processes are allowed to enter after a request
 - D) A limit on the system clock
-

18. Which condition of deadlock can be eliminated by requiring that a process requests all its required resources at once?

- A) Mutual exclusion
 - B) Hold and wait
 - C) No preemption
 - D) Circular wait
-

19. What is a binary semaphore commonly known as?

- A) Counter
 - B) Mutex
 - C) Switch
 - D) Gate
-

20. If a semaphore value is 0, what does it signify?

- A) The resource is available
 - B) The resource is busy
 - C) The process is finished
 - D) The system is idle
-

Answer Key — Operating System - Deadlocks & Synchronization

1. A	2. C	3. B	4. C	5. A
6. B	7. A	8. A	9. B	10. B
11. A	12. B	13. D	14. A	15. D
16. A	17. C	18. B	19. B	20. B

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Operating System - File Systems

1. What is the standard logical unit of information storage in an operating system?
 - A) Directory
 - B) File
 - C) Folder
 - D) Sector

2. Which file allocation method involves linking blocks of a file together via pointers?
 - A) Contiguous allocation
 - B) Linked allocation
 - C) Indexed allocation
 - D) Bit vector allocation

3. What is the primary purpose of a file directory?
 - A) To store file data
 - B) To format the disk
 - C) To organize and keep track of files
 - D) To increase CPU speed

4. In which allocation method is a file stored in a set of adjacent blocks on the disk?
 - A) Contiguous
 - B) Linked
 - C) Indexed
 - D) Hashed

5. What is the primary disadvantage of contiguous file allocation?
 - A) Slow access time
 - B) External fragmentation
 - C) High overhead
 - D) Inefficient storage

6. Which file system structure contains a list of all files and their attributes in a hierarchical manner?
 - A) FAT
 - B) Directory
 - C) Linked List
 - D) Stack

7. What is an 'absolute path' in a file system?
 - A) The path relative to the current folder
 - B) A path starting from the root directory
 - C) The name of the file only
 - D) A shortcut link

8. Which data structure is used to manage free disk space using 0s and 1s?
 - A) Linked list
 - B) Bit map
 - C) Grouping
 - D) Counting

9. What information is NOT typically found in a File Control Block (FCB)?
 - A) Permissions
 - B) File owner
 - C) Disk address
 - D) Current CPU process ID

10. What is a 'Mounting' operation in the context of file systems?

- A) Deleting a partition
 - B) Attaching a file system to the directory tree
 - C) Formatting a disk
 - D) Searching for a file
-

11. Which of the following is a common file system type used in Windows OS?

- A) ext4
 - B) NTFS
 - C) HFS+
 - D) UFS
-

12. In Indexed allocation, what structure holds the pointers to all the blocks of a file?

- A) Index block
 - B) File header
 - C) Allocation table
 - D) Directory entry
-

13. What is the name of the file system component that maintains the mapping of file names to their disk locations?

- A) Buffer cache
 - B) Directory
 - C) Scheduler
 - D) Kernel
-

14. What does the term 'File Descriptor' represent?

- A) A hardware address
 - B) An integer that uniquely identifies an open file
 - C) A storage sector
 - D) A file name extension
-

15. Which file access method reads records in order, one by one?

- A) Random access
 - B) Sequential access
 - C) Indexed access
 - D) Hashed access
-

16. Which system call is used to create a new file in Unix/Linux systems?

- A) open()
 - B) creat()
 - C) read()
 - D) write()
-

17. What is the benefit of using Indexed allocation?

- A) Supports direct access
 - B) No external fragmentation
 - C) No index block needed
 - D) Very high speed
-

18. A file 'extension' is primarily used to indicate:

- A) File size
 - B) File creation date
 - C) File type or format
 - D) Security level
-

19. Which disk space management technique uses a linked list of free blocks and stores the addresses of other free blocks in each free block?

- A) Bit vector
 - B) Linked list
 - C) Grouping
 - D) Counting
-

20. What happens when a file is 'deleted' in most operating systems?

- A) The data is instantly overwritten with zeros
 - B) The disk space is marked as free in the file system structure
 - C) The hardware is physically wiped
 - D) The file is moved to the CPU cache
-

Answer Key — Operating System - File Systems

1. B	2. B	3. C	4. A	5. B
6. B	7. B	8. B	9. D	10. B
11. B	12. A	13. B	14. B	15. B
16. B	17. A	18. C	19. C	20. B

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Operating System - OS Fundamentals & Types

1. Which of the following is considered the primary interface between the user and the computer hardware?
 - A) Application Software
 - B) Operating System
 - C) Compiler
 - D) Utility Software

2. An operating system that allows multiple users to work on the same system simultaneously is called?
 - A) Single-user OS
 - B) Multi-user OS
 - C) Batch OS
 - D) Real-time OS

3. Which of the following is a real-time operating system?
 - A) Windows 10
 - B) Linux
 - C) RTLinux
 - D) MS-DOS

4. What is the core part of an operating system that stays in memory during the entire computer session?
 - A) Shell
 - B) Kernel
 - C) BIOS
 - D) Driver

5. In a time-sharing system, the CPU switches between processes so quickly that it appears to the user that all programs are running:
 - A) Sequentially
 - B) Simultaneously
 - C) In background
 - D) In batch

6. Which type of operating system processes data as it arrives without any significant delay?
 - A) Batch OS
 - B) Real-time OS
 - C) Distributed OS
 - D) Network OS

7. A program in execution is formally known as:
 - A) Task
 - B) Process
 - C) Job
 - D) Thread

8. Which command is used to change the current directory in Unix/Linux systems?
 - A) cd
 - B) ls
 - C) mkdir
 - D) pwd

9. What is the process of booting the computer when it is already powered on?

- A) Cold boot
- B) Warm boot
- C) Hard boot
- D) Power-on self-test

10. Which of the following is NOT an operating system?

- A) Android
- B) Oracle
- C) Unix
- D) FreeBSD

11. In which OS architecture is the kernel kept as small as possible, moving most services to user space?

- A) Monolithic kernel
- B) Microkernel
- C) Hybrid kernel
- D) Exokernel

12. A system that connects multiple independent computers to work as a single system is a:

- A) Parallel OS
- B) Network OS
- C) Distributed OS
- D) Batch OS

13. What is the purpose of the 'Shell' in an operating system?

- A) To execute hardware instructions
- B) To interpret user commands
- C) To manage disk partitions
- D) To provide GUI exclusively

14. Which of the following operating systems is open-source?

- A) Windows
- B) macOS
- C) Linux
- D) Solaris

15. What is the primary function of a Scheduler in an OS?

- A) To manage files
- B) To manage CPU time
- C) To manage secondary memory
- D) To handle user inputs

16. The condition where a system becomes unresponsive because of high overhead is often called:

- A) Thrashing
- B) Deadlock
- C) Starvation
- D) Booting

17. What is the main characteristic of a Batch Operating System?

- A) User interaction
- B) Lack of user interaction
- C) High cost
- D) Real-time processing

18. Which of these is a feature of a multi-threaded process?

- A) Independent memory space
 - B) Sharing the same address space
 - C) No synchronization needed
 - D) Slower than single-thread
-

19. Which layer sits directly above the hardware in a layered operating system architecture?

- A) Application layer
 - B) User interface
 - C) CPU scheduler
 - D) Hardware abstraction layer
-

20. In an OS, what are the system calls used for?

- A) Hardware diagnostics
 - B) Interface between user programs and the kernel
 - C) Installing applications
 - D) Formatting disks
-

Answer Key — Operating System - OS Fundamentals & Types

1. B	2. B	3. C	4. B	5. B
6. B	7. B	8. A	9. B	10. B
11. B	12. C	13. B	14. C	15. B
16. A	17. B	18. B	19. D	20. B

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Networking - OSI/TCP-IP Model & Layers

1. Which layer of the OSI model is responsible for routing data packets between different networks?
 - A) Physical
 - B) Data Link
 - C) Network
 - D) Transport

2. In the OSI model, which layer ensures reliable end-to-end communication and error recovery?
 - A) Transport
 - B) Session
 - C) Presentation
 - D) Application

3. Which device operates primarily at the Data Link Layer (Layer 2) of the OSI model?
 - A) Router
 - B) Hub
 - C) Switch
 - D) Gateway

4. The TCP/IP model consists of how many layers?
 - A) 4
 - B) 5
 - C) 6
 - D) 7

5. Which PDU (Protocol Data Unit) is associated with the Data Link Layer?
 - A) Segment
 - B) Packet
 - C) Frame
 - D) Bit

6. Which layer is responsible for translating data into a format that the application layer can understand, such as encryption or compression?
 - A) Session
 - B) Presentation
 - C) Transport
 - D) Network

7. Which layer handles the physical transmission of raw bit streams over a medium?
 - A) Data Link
 - B) Physical
 - C) Network
 - D) Transport

8. The HTTP and FTP protocols operate at which layer of the OSI model?
 - A) Transport
 - B) Session
 - C) Presentation
 - D) Application

9. Which protocol is primarily associated with the Transport Layer?

- A) IP
 - B) TCP
 - C) Ethernet
 - D) ARP
-

10. What is the main function of the Session Layer in the OSI model?

- A) Addressing
 - B) Routing
 - C) Session management and synchronization
 - D) Bit transmission
-

11. Which layer uses logical addressing (IP addresses) to deliver data?

- A) Physical
 - B) Data Link
 - C) Network
 - D) Transport
-

12. In the TCP/IP suite, which layer corresponds to the OSI model's Physical and Data Link layers combined?

- A) Network Access
 - B) Internet
 - C) Transport
 - D) Application
-

13. A Hub operates at which layer of the OSI model?

- A) Physical
 - B) Data Link
 - C) Network
 - D) Transport
-

14. Which layer is responsible for flow control and error detection?

- A) Physical
 - B) Data Link
 - C) Network
 - D) Application
-

15. Which of the following is NOT a layer in the TCP/IP model?

- A) Application
 - B) Transport
 - C) Presentation
 - D) Internet
-

16. Which device is used to connect two different networks with different protocols?

- A) Switch
 - B) Hub
 - C) Gateway
 - D) Repeater
-

17. The process of adding control information to a packet as it moves down the OSI layers is called?

- A) Decapsulation
 - B) Encapsulation
 - C) Framing
 - D) Routing
-

18. MAC addresses are used at which layer of the OSI model?

- A) Physical
 - B) Data Link
 - C) Network
 - D) Transport
-

19. Which layer provides services directly to the user's software applications?

- A) Session
 - B) Presentation
 - C) Application
 - D) Transport
-

20. Which OSI layer defines the mechanical, electrical, and procedural interface for the physical medium?

- A) Physical
 - B) Data Link
 - C) Network
 - D) Transport
-

Answer Key — Networking - OSI/TCP-IP Model & Layers

1. C	2. A	3. C	4. A	5. C
6. B	7. B	8. D	9. B	10. C
11. C	12. A	13. A	14. B	15. C
16. C	17. B	18. B	19. C	20. A

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Networking - Protocols (HTTP, FTP, SMTP, DNS, DHCP)

1. Which protocol is primarily used for transferring files between a client and a server on a computer network?
 - A) HTTP
 - B) FTP
 - C) SMTP
 - D) DHCP

2. What is the primary function of the Domain Name System (DNS)?
 - A) Assigning IP addresses to hosts
 - B) Routing packets across networks
 - C) Translating domain names into IP addresses
 - D) Managing email transmission

3. Which port number is typically used by the HTTP protocol for unencrypted web traffic?
 - A) 21
 - B) 25
 - C) 80
 - D) 53

4. Which protocol is used by a client computer to automatically obtain an IP address from a server?
 - A) SMTP
 - B) DNS
 - C) DHCP
 - D) FTP

5. Which protocol is specifically designed for sending email messages?
 - A) POP3
 - B) SMTP
 - C) FTP
 - D) HTTP

6. What is the default port number used for secure HTTPS communication?
 - A) 80
 - B) 443
 - C) 21
 - D) 25

7. The process of mapping a human-readable website name to a numerical IP address is performed by which service?
 - A) DHCP
 - B) DNS
 - C) SMTP
 - D) FTP

8. Which protocol is categorized as a "connectionless" protocol because it uses UDP for its operations?
 - A) HTTP
 - B) FTP
 - C) DHCP
 - D) SMTP

9. Which of the following protocols is used for retrieving emails from a mail server?
 - A) POP3
 - B) FTP
 - C) DHCP
 - D) HTTP

10. In a DHCP process, what is the first message sent by a client to discover available servers?

- A) DHCPREQUEST
 - B) DHCPACK
 - C) DHCPOFFER
 - D) DHCPDISCOVER
-

11. Which protocol uses port 21 for its control connection?

- A) HTTP
 - B) SMTP
 - C) FTP
 - D) DNS
-

12. Which of the following is NOT an application layer protocol?

- A) HTTP
 - B) DNS
 - C) TCP
 - D) SMTP
-

13. Which protocol is used to provide a reliable, connection-oriented service for web browsing?

- A) TCP/HTTP
 - B) UDP/HTTP
 - C) DHCP
 - D) FTP
-

14. What does the acronym SMTP stand for?

- A) Simple Mail Transfer Protocol
 - B) Server Message Transfer Protocol
 - C) Simple Message Transmission Protocol
 - D) Secure Mail Transfer Protocol
-

15. Which protocol is used to synchronize the clocks of computers over a network?

- A) DNS
 - B) NTP
 - C) DHCP
 - D) FTP
-

16. Which of the following services relies on DNS to function correctly?

- A) File Transfer
 - B) Web Browsing
 - C) Dynamic IP Assignment
 - D) Email Delivery
-

17. If a computer is unable to resolve "www.google.com" to an IP address, which service is likely malfunctioning?

- A) DHCP
 - B) FTP
 - C) DNS
 - D) HTTP
-

18. Which protocol is used to access files on a remote server?

- A) HTTP
 - B) FTP
 - C) SMTP
 - D) DHCP
-

19. Which port number is used by the DNS service for queries?

- A) 53
 - B) 80
 - C) 25
 - D) 67
-

20. Which protocol is responsible for managing the dynamic configuration of IP addresses on a network?

- A) HTTP
 - B) DNS
 - C) DHCP
 - D) SMTP
-

Answer Key — Networking - Protocols (HTTP, FTP, SMTP, DNS, DHCP)

1. B	2. C	3. C	4. C	5. B
6. B	7. B	8. C	9. A	10. D
11. C	12. C	13. A	14. A	15. B
16. B	17. C	18. B	19. A	20. C

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Networking - Network Devices & Topologies

1. Which network topology requires a central controller or hub to which all nodes are connected?
 - A) Bus
 - B) Mesh
 - C) Star
 - D) Ring

2. In a bus topology, what is the name of the central cable to which all nodes are connected?
 - A) Backbone
 - B) Segment
 - C) Trunk
 - D) Bus line

3. Which networking device is often referred to as a "multi-port repeater"?
 - A) Router
 - B) Switch
 - C) Bridge
 - D) Hub

4. Which topology provides the highest level of redundancy, where every device is connected to every other device?
 - A) Star
 - B) Full Mesh
 - C) Tree
 - D) Hybrid

5. What is the primary disadvantage of a bus topology?
 - A) High cost
 - B) Complex setup
 - C) Failure of the backbone cable brings down the entire network
 - D) Difficult to add new nodes

6. Which device is capable of filtering traffic based on MAC addresses?
 - A) Hub
 - B) Switch
 - C) Repeater
 - D) Modem

7. A network topology where nodes are connected in a closed loop is known as?
 - A) Star
 - B) Ring
 - C) Bus
 - D) Mesh

8. Which device is used to connect two LAN segments that use the same protocol?
 - A) Router
 - B) Gateway
 - C) Bridge
 - D) Repeater

9. In a ring topology, what is the name of the small data packet that circulates to grant a node permission to transmit?
 - A) Token
 - B) Frame
 - C) Signal
 - D) Packet

10. Which topology is a combination of two or more different topologies?

- A) Star
- B) Tree
- C) Hybrid
- D) Mesh

11. Which device operates at the Network Layer to determine the best path for data packets?

- A) Router
- B) Switch
- C) Hub
- D) NIC

12. What is the main purpose of a repeater in a network?

- A) To switch data packets
- B) To connect different networks
- C) To regenerate and extend signals
- D) To filter traffic

13. A tree topology is essentially a variation of which other topology?

- A) Bus
- B) Ring
- C) Star
- D) Mesh

14. Which hardware component is required to connect a computer to a physical network medium?

- A) Router
- B) NIC
- C) Bridge
- D) Gateway

15. Which network device is considered "intelligent" because it learns the MAC addresses of connected devices?

- A) Hub
- B) Switch
- C) Repeater
- D) Modem

16. In which topology does the failure of one node disrupt the communication of the entire network?

- A) Star
- B) Bus
- C) Mesh
- D) Hybrid

17. What is the main difference between a hub and a switch?

- A) Hubs are faster
- B) Hubs broadcast data to all ports, while switches send data to specific ports
- C) Switches only work with fiber optics
- D) Hubs support VLANs

18. Which network device functions as an entry/exit point of a network, often connecting to the Internet?

- A) Hub
- B) Gateway
- C) Repeater
- D) NIC